Experiment 03

Experiments on CISCO PACKET TRACER (Simulation Tool)

Aim

To understand Packet Tracer, design simple networks and analyse hub and switch behaviour.

Requirements

Cisco Packet Tracer; four to six PCs; Hub-PT; 2960 switch; Ethernet connections.

Procedure / algorithm

1. Install/open Packet Tracer. Identify the device-selection box, Logical/Physical workspace, common tools, Realtime/Simulation tabs and event/PDU list.

2. Create four PCs and a hub. Connect each host to a hub port using copper straight-through cables.

3. Assign the unique 192.168.1.1-4/24 addresses in addressing.csv. No default gateway is needed within this single subnet.

4. Send a Simple PDU between hosts. In Simulation mode inspect ARP and ICMP events and the ports receiving the repeated traffic.

5. Build the same network with a switch and repeat. Inspect show mac address-table before and after communication.

6. Build the mixed topology with two PCs on the hub and two on the switch; use an uplink or automatic cable choice for the hub-switch connection.

7. Compare forwarding and collision domains. Inspect the actual college network and draw its observed labelled topology separately.

Complete source / configuration: addressing.csv

device	ip	mask	gateway
PC1	192.168.1.1	255.255.255.0	none
PC2	192.168.1.2	255.255.255.0	none
PC3	192.168.1.3	255.255.255.0	none
PC4	192.168.1.4	255.255.255.0	none

Complete source / configuration: switch.cfg

```
enable
configure terminal
hostname SW1
interface range FastEthernet0/1-4
 switchport mode access
 no shutdown
end
show mac address-table
show interfaces status
```

Diagram: mixed

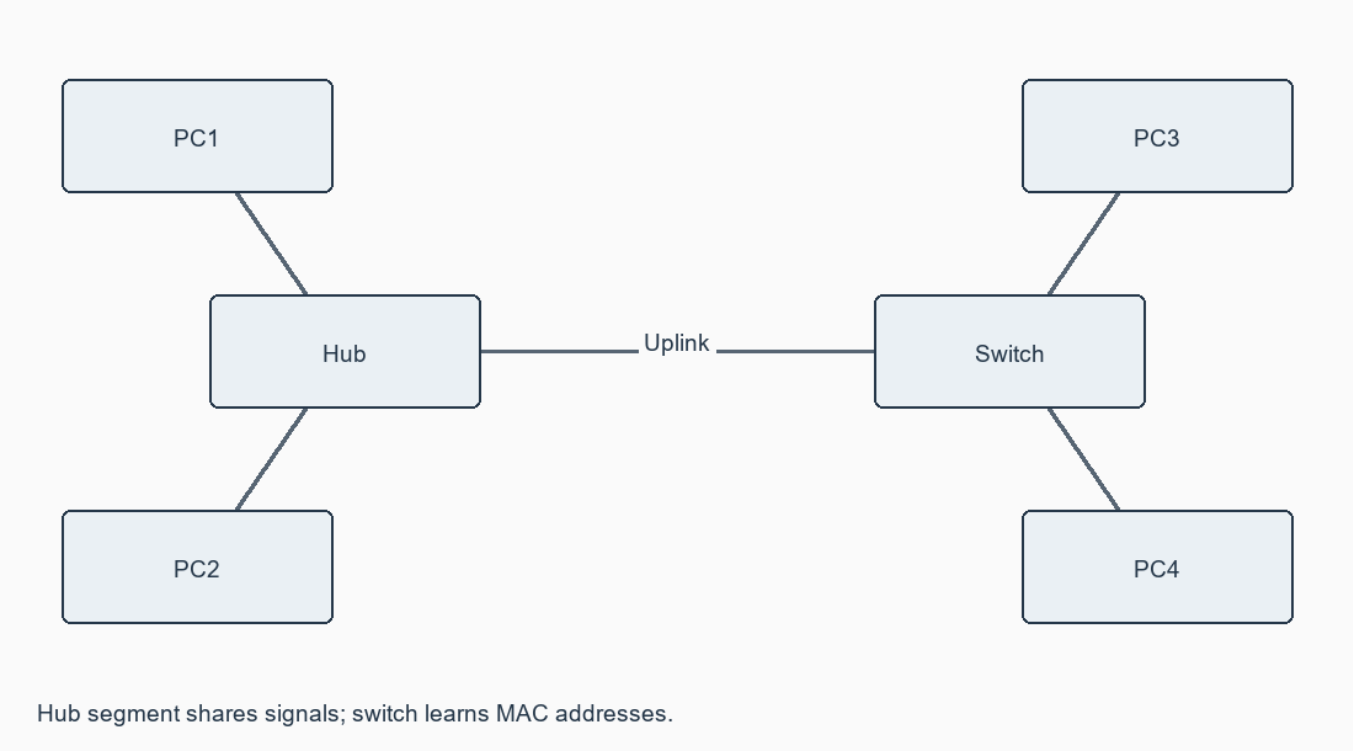
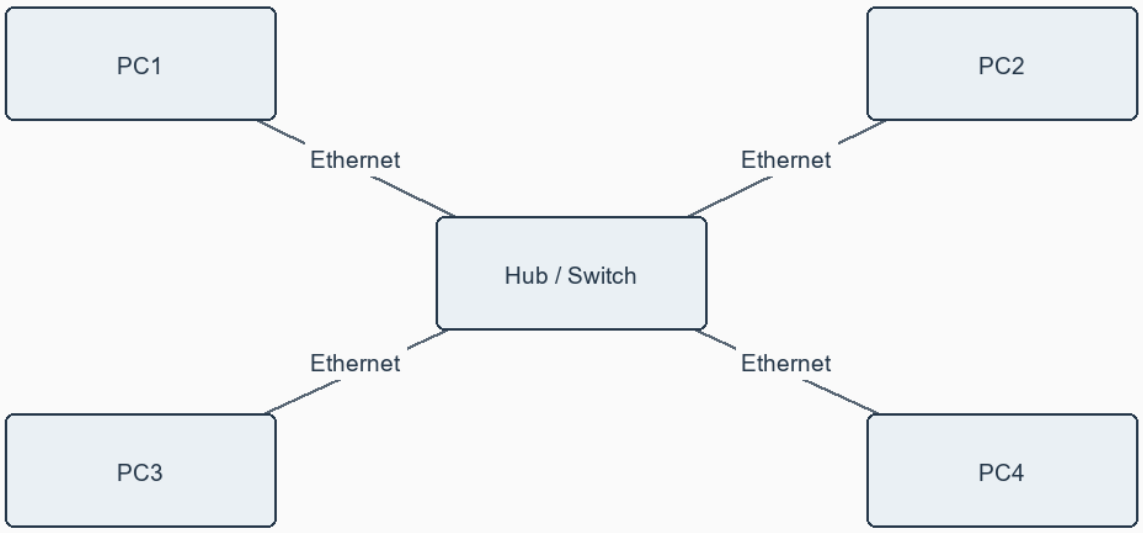


Diagram: star



Physical star: each host connects to a central port.

Actual output / verification

Not executed in the required lab/tool environment. No actual operational output is claimed.

Expected output - not an execution record

All four assigned hosts communicate. Hub-connected hosts see repeated signals; switch unicast forwarding becomes selective after MAC learning. ARP broadcasts are flooded.

Observations

A hub repeats signals to all other ports and has one shared collision domain. A switch learns source MAC-to-port mappings and forwards known unicast to the required port. Broadcasts and unknown unicast are flooded within the VLAN. An actual college topology is not invented.

Result

Hub-only, switch-only and combined designs, addresses and switch commands are supplied. Packet Tracer execution and campus inspection remain pending.